

Heaven's light is Our Guide
Rajshahi University of Engineering & Technology (RUET)
Department of Computer Science & Engineering
Course Title: Digital Logic Design, Course Code: CSE 2103
Assignment, Total Marks: 10, Submission Deadline: 30/09/2026

Topic: Design and Analysis of a Cascaded MOD-100 Hall Occupancy Counter (Asynchronous vs. Synchronous Design) **CO3-PO3-C4 (Engineering Analysis)**

Scenario

RUET's newly built 250-seat "Innovation Hall" needs a digital occupancy display at its entrance, showing the number of people currently inside as a two-digit decimal number (00 to 99), using a turnstile sensor that pulses once per person entering. The display is built from two cascaded decade (MOD-10) counters — a units-digit stage and a tens-digit stage — together forming a MOD-100 counter. Under normal footfall the turnstile pulses slowly, but during rush periods (e.g., just before an exam or seminar begins), the sensor can generate a rapid burst of pulses at up to 500 kHz as students crowd through. The hall's technician reports that, during such rush bursts, the displayed occupancy count occasionally jumps or briefly flickers to an incorrect value before settling, even though the display always eventually shows the correct count once the crowd clears. Your class has been asked to analyze the existing design and propose a verified redesign so this no longer happens.

Tasks

- (a) Counter Architecture: Determine the number of flip-flops needed to build each MOD-10 decade stage, and the total number of flip-flops for the complete MOD-100 (00-99) occupancy counter using two cascaded decade stages. Briefly describe how the tens-digit stage should be clocked from the units-digit stage so the two decades count correctly together.
- (b) Asynchronous (Ripple) Analysis — Existing Design: The existing decade counters are built as asynchronous (ripple) counters using J-K flip-flops, each with a propagation delay of 10 ns. (i) Calculate the worst-case cumulative propagation delay across all flip-flops involved when a carry must ripple from the units stage into the tens stage. (ii) Determine the maximum safe operating frequency, F_{max} , for this ripple design. (iii) Analyze, with justification, whether operating at the turnstile's 500 kHz rush-burst rate can explain the occasional flicker/incorrect-count fault reported by the technician.
- (c) Synchronous Redesign — Single Decade Stage: Design ONE synchronous MOD-10 (BCD) decade counter using J-K flip-flops. Your report must include: the complete state transition diagram (0 through 9, showing the $9 \rightarrow 0$ rollover and any unused states 10-15); the PRESENT-state/NEXT-state excitation table; and the simplified J and K input expressions for all four flip-flops.
- (d) System-Level Comparison: (i) Explain how the tens-digit stage can be enabled synchronously (rather than clocked asynchronously by the units stage) using a terminal-count/enable signal derived from the units stage, so that both decades ultimately share the same clock. (ii) Estimate the new F_{max} for this fully synchronous MOD-100 design (state any assumed flip-flop and logic-gate delays), and compare it against the ripple design's F_{max} from part (b).
- (e) Critical Reflection: In 4-6 sentences, compare (1) a fully asynchronous cascade, (2) a fully synchronous design, and (3) a hybrid design (synchronous within each decade, ripple-carry between decades) for this hall-counter application, in terms of speed, hardware/gate complexity, and reliability, and recommend which one you would deploy and why.

Submission Guidelines

- Submit as a single typed PDF report with clearly labeled sections (a)-(e).
- All diagrams (block/cascade diagram, state transition diagram) must be neatly drawn/typed and labeled.
- Show all numerical calculations step-by-step; unsupported final answers will not receive full credit.

Assessment Rubric (Total: 10 Marks)

Assessment Criterion	CO / PO / Bloom's Level Mapped	Marks Allocated
(a) Correct flip-flop count and cascade-clocking description	CO3-PO3 / C4-Analyze	2
(b) Correct cumulative-delay/Fmax calculation and sound timing-fault analysis	CO3-PO3 / C4-Analyze	2
(c) Correct synchronous decade-counter design (state diagram, excitation table, J-K equations)	CO3-PO3 / C4-Analyze	3
(d) Correct synchronous enable/cascade explanation and Fmax comparison	CO3-PO3 / C4-Analyze	2
(e) Depth and soundness of the three-way critical comparison and recommendation	CO3-PO3 / C4-Analyze	1

Detailed Performance-Level Rubric

Criterion	Excellent	Good	Fair	Poor
Architecture Understanding (FF count & cascading)	Correct FF count and a clear, correct explanation of inter-stage clocking.	Correct with minor gaps in the cascading explanation.	Partially correct; cascading logic misunderstood.	Incorrect or missing.
Timing Analysis (Ripple Delay, Fmax, Fault Diagnosis)	All calculations correct with clear reasoning; fault diagnosis technically sound and well justified.	Minor calculation slips but correct overall diagnosis.	Formulae correct but weak/incorrect application to the fault scenario.	Incorrect or missing calculations/diagnosis.
Synchronous Design Correctness	State diagram, excitation table, and J-K equations all complete and correct.	Correct with minor errors in one element.	Partially correct; method understood but notable errors present.	Design incorrect, incomplete, or missing.
Critical Comparison & Recommendation	Insightful, well-justified three-way comparison with a clear, sound recommendation.	Sound comparison with adequate justification.	Basic comparison with limited justification.	Comparison superficial, generic, or absent.